MILITARY SPECIFICATION

CASE, CARTRIDGE, M205E1, 152MM, PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.-This specification covers the component parts of the nonmetallic cartridge case designated as Case, Cartridge Assembly 152MM, M205E1 (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard, Quality Assurance Provisions, General Specification For
MIL-C-50534 - Case, Cartridge, M205, 152MM, Parts For

FEDERAL

PPP-B- 26 - Bag, Plastic, Polyethylene

DRAWINGS

ARMY

9275739 - Case, Cartridge, Assembly, 152MM, M205E1
9289037 - Cartridge, 152MM, TP-T, M411A3E1
8889489 - Box Fiber, for Body for Case, Cartridge, M157, M205 or M205E1
9242503 - Box, Fiber, for Base, for Cartridge Case, M157, M205 or M205E1
8799112 - Marking Diagram and Sealing of Ammunition Fiberboard Containers

FSC: 1320

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CODE OF FEDERAL REGULATIONS

Title 49, Transportation: Parts 100-199

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the above publication should cite: 49 CFR 100-199 (latest revision).

3. REQUIREMENTS

3.1 Materials.-The constituent materials shall comply with the requirements of all applicable drawings and specifications.

3.2 Parts.-The cartridge case parts shall comply with all the requirements specified in Drawing 9275739 and with all the requirements specified in MIL-C-50534.

3.3 Process Data.-The following process data shall be recorded and furnished to the Government in accordance with data item description DI-P-1604 as specified in the contract (see 6.4).

3.3.1 Solution composition for each batch of polyvinyl alcohol (PVA), Saran and silicone/fungicide used.

3.3.2 Temperature, viscosity and flowrate of PVA and Saran solutions each hour of operation.

3.3.3 Atomization pressure, line speed and spindle speed each hour of operation and whenever a change is made.

3.3.4 For each sprayhead used, record manufacturer and model number, vertical height from base of spindle, horizontal distance from center of spindle and angle to horizontal.

3.3.5 For each solution applied, record spray time, dry time, dry temperature and humidity each hour, cure time (as applicable) and cure temperature each hour (as applicable).

3.3.6 Silicone/fungicide solution temperature, viscosity, diptime, dry time and dry temperature.

3.3.7 Coating weight for each coating and impregnating solution (see 4.4.2.3).

3.3.8 Coating Uniformity (see 4.4.2.7).

3.4 Case Part Weight.-The M205E1 cartridge case parts shall comply with the requirements of Table I when tested in accordance with 4.4.2.4.

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TABLE I
CASE PART WEIGHT

Item	Weight	Tolerance
Body	302.3 gms	± 12.0 gms
Base	76.4 gms	± 4.0 gms

3.5. Workmanship.-All parts shall be fabricated in a thorough, workmanlike manner. Each case part shall be visually examined after each coating operation. Any evidence of cracks, pores, discontinuities, runs, sags, blisters, pinholes, foreign material or damage to any part shall be reported. Each part shall be examined after each coating application to verify that inside lip area of base and outside scalloped area of body are adequately protected from contact with coating. Any evidence of insufficient protection shall be reported.

3.6 Proving Ground.-The cartridge case, when assembled as part of a 152MM Cartridge, shall function satisfactorily without smoldering residue, and without more than one occurrence of excessive residue (see 6.2) when tested in accordance with 4.4.2.5.

3.7 Moisture-Vapor Transmission.-The moisture pickup of the case assembly shall be less than 1.5% when tested in accordance with 4.4.2.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

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4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Case, Cartridge Assembly 152MM, M205E1	9275739	120

4.3.2 Inspections to be Performed.-The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and to the inspections of Table II. All inspections listed in Table II shall be classified Major unless otherwise specified.

4.3.3 Rejection.-See MIL-A-48078 (PA).

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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TABLE II - FIRST ARTICLE INSPECTION

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48884 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Case, Cartridge, M205E1, 152MM, Parts for				9275739
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Physical Properties	30		3.4	4.4.2.4
A	Proving Ground	40		3.6	4.4.2.5
	Moisture Vapor Transmission	10		3.7	4.4.2.6

NOTES:

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QUALITY CONFORMANCE INSPECTION

MIL-C-48884 (PA

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299330
4.4.2.1	Body for Case Cartridge, M205E1				NEXT HIGHER ASSEMBLY 9275739
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor	Workmanship		100%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48884 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Base, For Case Cartridge, M205E1				9299329
					NEXT HIGHER ASSEMBLY
					9275739
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor	Workmanship		100%	3.5	Visual

NOTES:

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4.4.2.3 Coating Weight (see 3.3.7).-Thirty (30) case bodies and 30 case bases shall be selected prior to processing and tested for information in accordance with 4.5.1. These parts shall be coated in the following order: Five bodies and bases in the beginning of production and one body and base periodically until the end of production (the periods shall be evenly spaced based on production quantity). In addition, whenever any change is made in the process, an additional five (5) bodies or bases (as applicable) shall be tested as specified for information.

4.4.2.4 Case Part Weight (see 3.4).-Thirty (30) case bodies and 30 case bases shall be randomly selected from the lot. Each part shall be weighed to the nearest 0.1 gram. If any part fails the applicable requirement, all remaining parts in the lot shall be weighed. Only acceptable parts may be retained in the lot.

4.4.2.5 Proving Ground (see 3.6).-Forty (40) case bodies and 40 case bases shall be randomly selected from the lot and tested ballistically for smoldering and excessive residue at 70°F in accordance with 4.5.4. Failure to comply with 3.6 shall be cause to reject the lot.

4.4.2.6 Moisture Vapor Transmission (see 3.7).-Ten (10) case bodies and 10 case bases shall be randomly selected from the lot and tested in accordance with 4.5.3. If any part fails the requirement, the lot shall be rejected.

4.4.2.7 Coating Uniformity (see 3.3.8).-Five bodies and five bases shall be randomly selected from the lot and examined for information in accordance with 4.5.2.

4.5 Test Methods and Procedures

4.5.1 Coating Weight.-Each body and base shall be weighed to the nearest 0.1 gram before and after each coating and each impregnating operation. Samples shall be dried to constant weight at 122°F for each determination. Calculate the weight increase of each spray coating and impregnating solution by subtracting the weight before coating from the weight after coating. All data shall be tabulated and identified by part number to enable correlation of process data with subsequent laboratory test data. All parts shall be retained for laboratory testing as designated in 4.5.2 and 4.5.3.

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4.5.2 Coating Uniformity.-Four specimens, 1 inch by 1 inch, shall be cut from the sidewall of each body 90° apart (or from the flat of the base as applicable). A permanent polaroid photo shall be made of each specimen as seen under a scanning electron microscope at 100 magnification.

4.5.3 Moisture Vapor Transmission.-The bodies and bases shall be assembled with 2 kilograms of predried Indicating Grade Drierite (manufactured by the Hammond Corporation) or equivalent contained in the assembly. The base igniter boss, the interface between the body and the base, and the forward end of the body shall be sealed with wax. The assembly shall be placed in a conditioning chamber and exposed to 95% relative humidity at 100°F for 24 hours. The assembly of case and drierite then shall be weighed. The moisture pickup of the Drierite shall be calculated as follows:

$$\% \text{ Moisture} = \frac{100\ (B - A)}{A}$$

Where:
A = weight of case assembly and drierite before test.
B = weight of case assembly and drierite after test.

4.5.4 Proving Ground.-These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. Case parts shall be assembled to 152MM projectiles in accordance with dwg. 9289037. Without using adhesive to assemble the body to the base of the case. Rounds shall be conditioned for 24 hours at 70 ± 2°F. Rounds shall be ballistically tested within 5 minutes after removal from conditioning. The gun scavenger shall be used during the ballistic test. Observation shall be made for compliance with the applicable requirement (see 3.6). Movie camera coverage of the breech block and chamber of the weapon shall be provided for all test firings.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level C.-Each case body shall be sealed in a one and one half mil (minimum thickness) polyethylene bag. A maximum of twenty (20) case bases shall be packed together in a similar polyethylene bag. The bag material shall be Type II, Grade A, Style 1 in accordance with PPP-B-26. The method of sealing shall be a Government approved heat seal.

5.2 Packing

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5.2.1 Level C.-Bagged case bodies and case bases shall be packed in accordance with dwg. 8889489 or 9242503, as applicable.

5.3 Marking.-Unless otherwise specified, markings shall be in accordance with dwg. 8799112, and dwg. 9275739.

6. NOTES

6.1 Ordering Data.-Ordering data shall be in accordance with MIL-A-48078 (PA).

6.2 Excessive residue.-Residue is defined as excessive if it is significant enough to prevent chambering or hinder functioning of a subsequent round, or exceeds 1 inch by 2 inches in size, as determined by the proving ground.

6.3 Proving Ground

6.3.1 Assembly of cartridge case.-Cartridge cases shall be assembled to 152MM, XM411 or equivalent projectiles using normal loading procedures, except that no adhesive or lacquer shall be used to cement the case body to the case base, or in the assembly of the Primer to the case base.

6.3.2 Cleaning.-Prior to each firing, the breech and firing probe of the test weapon shall be thoroughly cleaned and dried.

6.3.3 Handling and storage.-Special care shall be taken in the handling and storage of sample cartridge cases to prevent contamination by grease, grime, dirt, or other foreign material.

6.4 Data Requirements.-Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
3.3	Process Data	DI-P-1604 Except only process data stipulated herein applies and report format optional

(Copies of data item descriptions required by the contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer. Distribution of data required by DI-P-1604 shall include Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801).

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6.5 The M205E1 Cartridge Case consists of a M205 Cartridge Case to which a water and moisture repellent coating system has been applied. This system consists of coating the case with polyvinyl alcohol, then coating with Saran and then impregnation of the case with a silicone/fungicide solution.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1320-A297

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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